

Part 1: Final Database Description.

R1.1

1.0 Project Overview and Intent

This library management database is designed to provide a centralized system for small community libraries to track inventory, borrowers' history, and asset availability. The database enables librarians to manage their catalog of approximately 1,000 books and assist patrons in finding available resources through real-time updates on book availability, categorical indexing, and overdue statuses.

2.0 Database Entities and Attributes

The database system consists of five core entities:

- **Books:** A physical or digital library item identified by a unique ISBN. Each book has a title, publication date, and a current copy count.
- **Patrons:** An authorized library member. Each patron has a unique ID, name, email address, and secure password for system access.
- **Authors:** The writer or co-author of one or more books. Each author has a unique ID and a name.
- **Genres:** Categorical classification for the collection of books. Each genre has a unique ID and a name.
- **System Logs:** A record of reported discrepancies or indexing errors flagged by users. Each log entry has a unique ID, a date, a text description, and an error_type classification.

3.0 Complexity and Relationship Logic

Three N:N Relationships:

1. Books and Authors (Authorship): One book can have many authors (co-authors), and one author can write many different books.
2. Books and Genres (Categorization): A single book can be tagged with multiple genres, and a genre contains many different books.
3. Patrons and Books (Circulation): One patron can sign out many books at once, and a specific book can be borrowed by many different patrons.

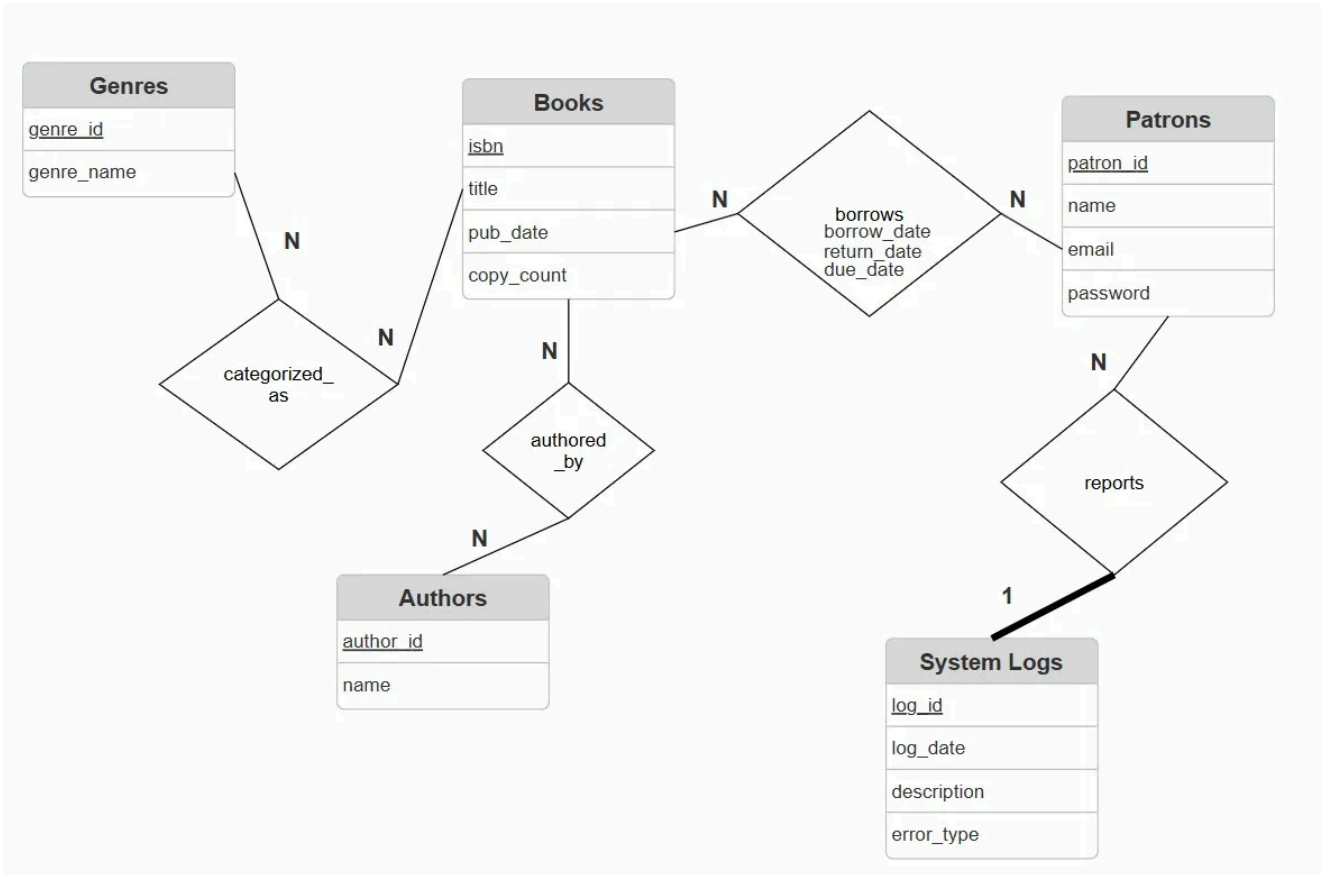
One 1:N Relationships:

1. Patrons and System Logs (Reporting): One patron can report many indexing errors or discrepancies over time, but each log entry is associated with exactly one patron.

4.0 Design Assumptions

1. A book represents the physical or digital library item with a title and an ISBN.
2. An author represents someone who has written or co-authored a book.
3. A specific book can have multiple authors.
4. An author can be associated with many different books.
5. A patron represents a community library member authorized to borrow resources.
6. A genre represents a categorical subject or classification for a book.
7. A patron can sign out many books at once and over time.
8. A specific book can be borrowed by many different patrons.
9. The system tracks the specific date a book was signed out, its expected due date, and the actual date it was returned to, to manage circulation history.
10. A book can be tagged with one or more genres.
11. The database stores unique credentials for patrons.
12. The system log represents a history of reported discrepancies or indexing errors flagged by users.
13. Each system log entry is uniquely linked to the individual patron who reported the indexing error or discrepancy.

R1.2-1.4



R1.5

